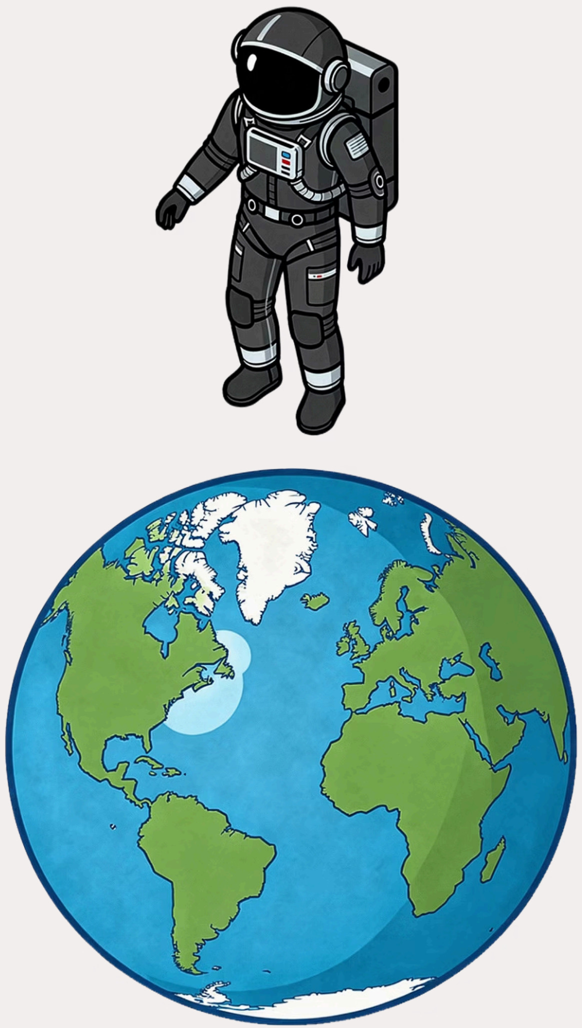


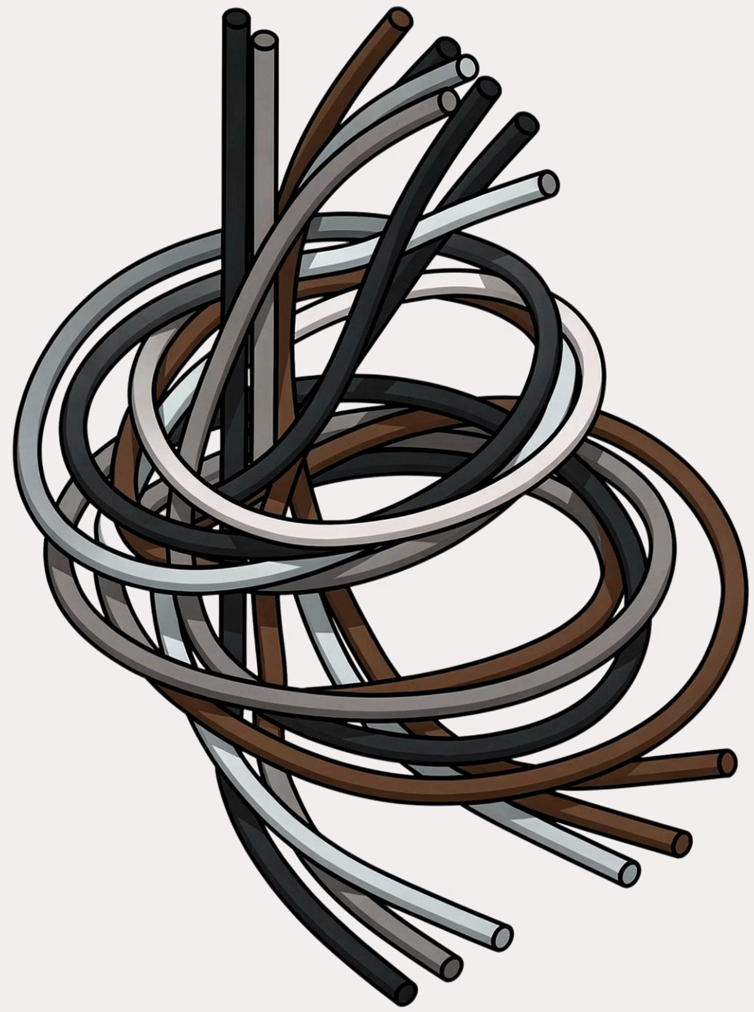
Astronomer Managed Airflow: Enterprise Orchestration, Simplified

A comprehensive guide to running Apache Airflow at enterprise scale — without the operational burden.

CHAPTER 1

The Enterprise Challenge in Data Orchestration

As data pipelines grow in complexity and scale, enterprises face mounting pressure to maintain reliability, security, and governance — all while keeping engineering teams focused on delivering business value rather than managing infrastructure.



Why Astronomer for Enterprise?

The Airflow Complexity Problem

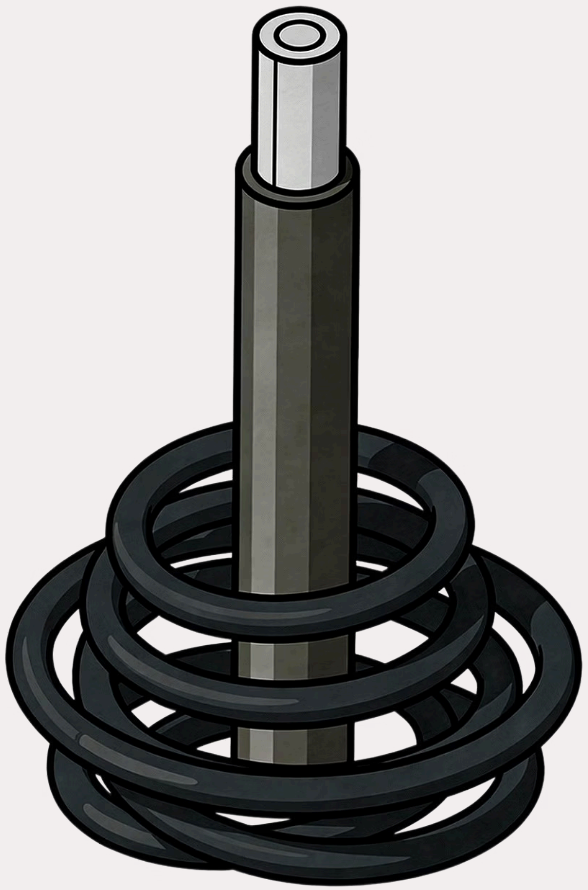
Managing Airflow at scale introduces significant infrastructure overhead, security concerns, and operational burdens. Kubernetes configuration, executor tuning, and metadata database management consume valuable engineering time that should be spent on business logic.

The Need for Control & Scale

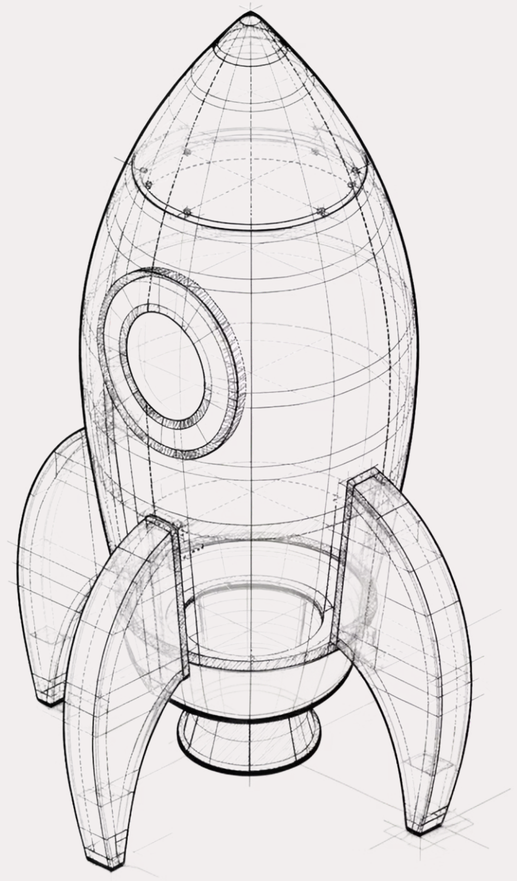
Enterprises require robust governance, security guardrails, and the ability to scale workflows without compromising performance or compliance. Ad-hoc Airflow deployments cannot meet these standards consistently.

Focus on Data, Not Infrastructure

Astronomer empowers teams to concentrate on building data pipelines and driving business value — not managing Kubernetes clusters, Airflow versions, or complex configurations.



Astronomer transforms Airflow from an operational challenge into a strategic business enabler — cutting through complexity with a single, managed platform.



CHAPTER 2

Understanding the Managed Architecture

Astro's architecture is built on a principle of clear separation: Astronomer manages the platform, so your team owns the pipelines. This chapter unpacks how the control plane, data plane, and deployment isolation work together.

Astro: A Fully-Managed SaaS for Data Orchestration

What is Astro?

Astro is Astronomer's fully-managed cloud platform that simplifies and optimizes Apache Airflow. It manages all underlying infrastructure, from provisioning to upgrades, so your team can focus entirely on authoring and running DAGs.

Key Platform Components

- **Astro Deployments** — isolated Airflow environments per project or team
- **Astro UI** — centralized dashboard for monitoring and management
- **Astro CLI** — developer-friendly local development and CI/CD tooling
- **Astro API** — programmatic platform control
- **Astro Hypervisor** — the intelligent autoscaling engine

Each Deployment runs in its own isolated Kubernetes namespace for security and stability.

Control Plane vs. Data Plane: A Clear Separation

Astro's architecture draws a deliberate boundary between orchestration and execution — giving enterprises flexibility without sacrificing control.

Control Plane

Astronomer-Managed

- Airflow Scheduler & Web Server
- Metadata database management
- Observability & alerting
- DAG versioning & deployment
- Platform-level security & auth

Data Plane

Customer-Controlled Options

- Actual task execution & compute
- Data processing workloads
- Hosted: Astronomer manages execution
- Remote: Your own Kubernetes infrastructure
- Supports cloud or on-premises environments

Hosted Execution: Effortless Infrastructure Management

How It Works

In Hosted Execution mode, Astronomer manages the full task execution infrastructure end-to-end. This includes autoscaling workers, provisioning compute, and managing the full worker lifecycle — all without any action required from your team.

Astronomer's Hypervisor monitors queue depth and resource utilization in real time, spinning workers up and down dynamically to match demand.

Ideal For

Cloud-Native Teams

Teams comfortable running workloads in a managed SaaS environment with network-accessible data sources.

Rapid Onboarding

Organizations prioritizing speed-to-value with minimal infrastructure setup and maintenance overhead.

Remote Execution: Your Infrastructure, Our Orchestration

Enterprise-Grade Security

Tasks execute entirely within your private infrastructure — cloud VPC or on-premises. Sensitive data never traverses Astronomer's managed environment, satisfying the strictest compliance requirements.

Bring Your Kubernetes

Point Astro at your existing Kubernetes cluster. Tasks run in your environment while Astro's control plane handles scheduling, observability, and all orchestration decisions.

Compliance & Data Residency

Meet data residency regulations, sovereignty requirements, and internal audit mandates. Ideal for financial services, healthcare, and government workloads with strict data localization rules.

Hybrid Flexibility

Seamlessly support hybrid and multi-cloud architectures. Run some workloads hosted, others remotely — all managed through a single unified control plane.

Remote Execution is available on Astronomer Enterprise tiers, enabling a clean separation of orchestration from execution.

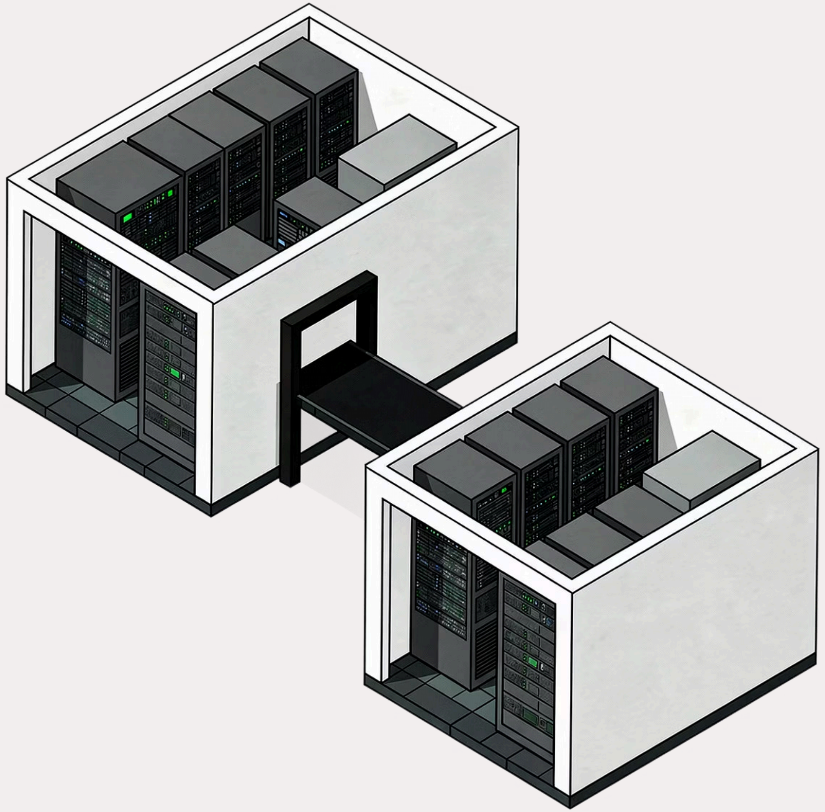
Orchestration Plane vs. Execution Plane

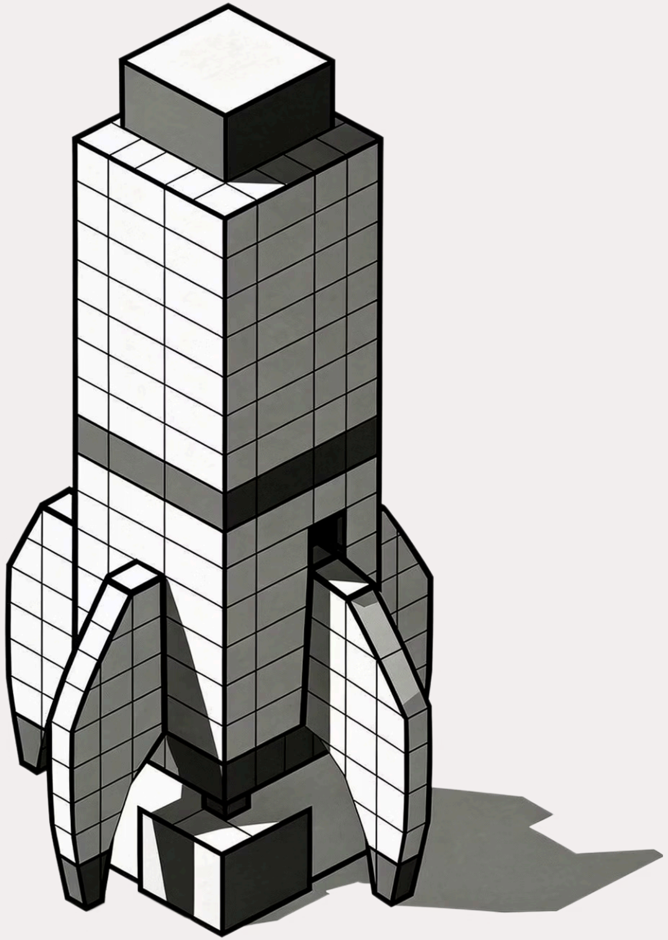
Orchestration Plane

Cloud-based, Astronomer-managed. Hosts the Scheduler, Web Server, metadata DB, and observability stack. Fully abstracted from your team's operational responsibility.

Execution Plane

Customer's own infrastructure — cloud or on-premises Kubernetes. Where tasks actually run, data is processed, and compute resources are consumed. Fully under your control.





CHAPTER 3

Deployment and Scaling Models

Astro's deployment architecture gives enterprises the isolation, flexibility, and scaling intelligence needed to run mission-critical workflows at any volume — without managing the underlying complexity.

Deployment Architecture: Flexible and Isolated

Astro organizes Airflow environments through a hierarchy of **Workspaces** and **Deployments**, providing both team-level autonomy and platform-wide governance.

1

Deployments

Individual Airflow environments running in isolated Kubernetes namespaces. Each Deployment has its own Scheduler, Worker pool, and configuration — ensuring workload isolation and stability.

2

Workspaces

Organizational constructs that group related Deployments. Teams manage access, roles, and permissions at the Workspace level, enabling self-service without compromising security boundaries.

3

Standard Clusters

Multi-tenant, cost-efficient clusters with isolated namespaces per Deployment. The default option for most teams — fast to provision and fully managed by Astronomer.

4

Dedicated Clusters

Single-tenant clusters for enhanced networking and security. Support VPC peering, AWS PrivateLink, and custom network policies — ideal for regulated industries.

Worker Scaling Model: Dynamic and Efficient

Autoscaling Intelligence

Astro's Hypervisor continuously monitors DAG queue depth and resource utilization, dynamically scaling compute resources to match real-time workload demands. Workers spin up instantly at peak and scale to zero during idle periods — optimizing both performance and cost.

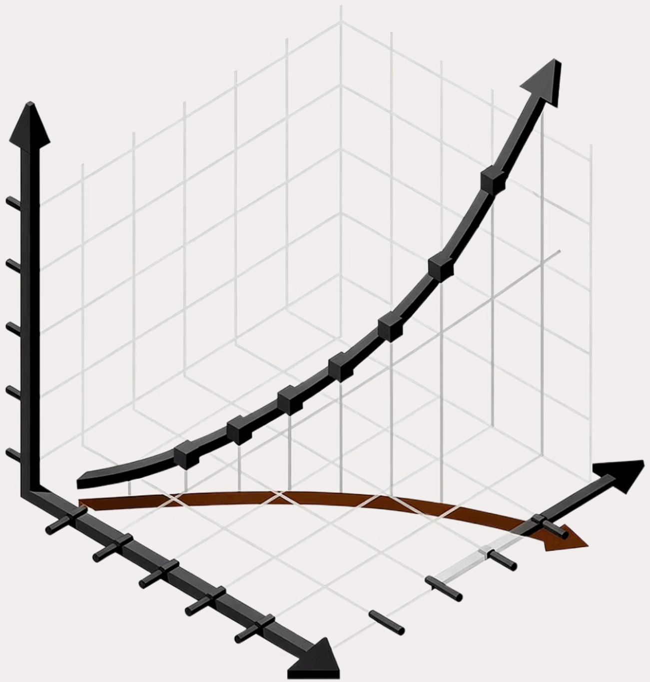
Executor Options & Resource Control

- **Local Executor** — ideal for development and lightweight workloads
- **Celery Executor** — distributed task execution for production at scale
- **Kubernetes Executor** — per-task pod isolation for maximum flexibility

Configure CPU, memory, and worker counts per Deployment to strike the right balance between performance and infrastructure cost.

Scale to zero during off-hours to eliminate idle compute costs entirely.

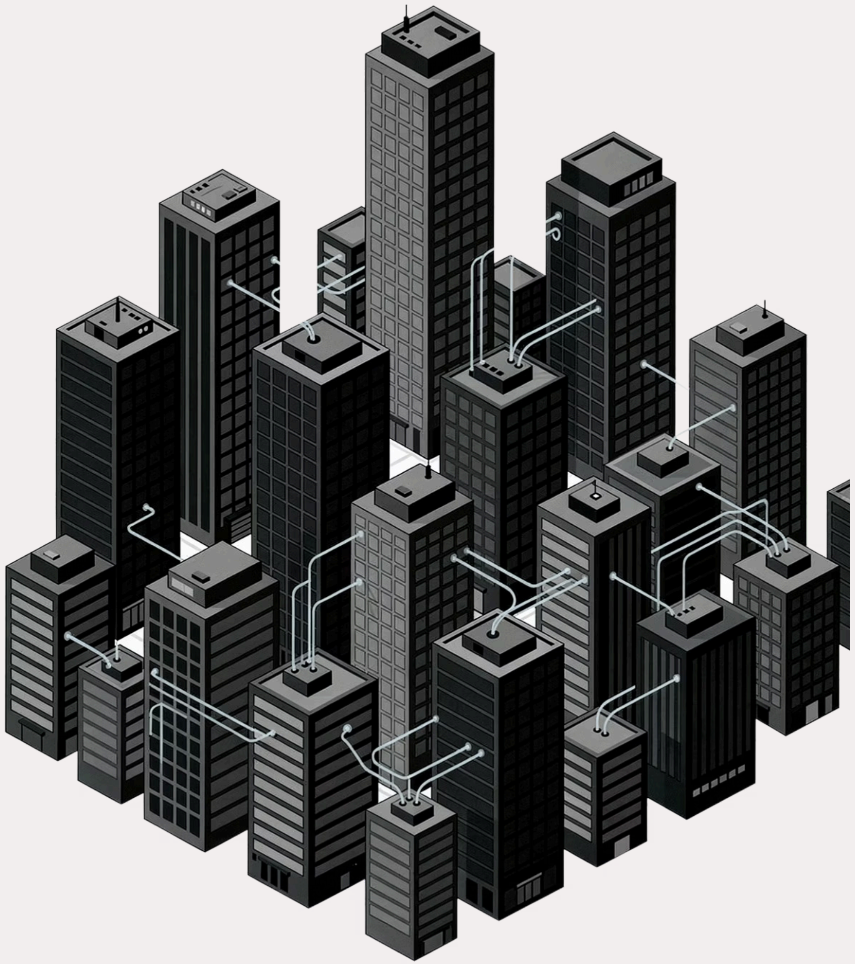
The Astro Hypervisor eliminates the guesswork of capacity planning — workers scale fluidly with your workload, ensuring pipelines never wait for compute and resources are never wasted.



CHAPTER 4

Enterprise Orchestration Use Cases

From financial services compliance to machine learning pipelines, Astronomer's managed platform addresses the real-world orchestration challenges that enterprise data teams face every day.



Use Case 1: Financial Services Compliance

The Challenge

Financial institutions face strict regulatory requirements including SOX, PCI-DSS, and GDPR. Data sovereignty mandates that sensitive financial data remain within defined geographic and network boundaries. Every pipeline execution must produce a verifiable audit trail.

The Astronomer Solution

Remote Execution ensures sensitive financial data never leaves the organization's private cloud or on-premises infrastructure. Astro's control plane manages orchestration and generates compliance reporting workflows, while dedicated clusters with PrivateLink support eliminate public internet exposure entirely.

Data Sovereignty

Tasks execute within your private network boundary — data never transits Astronomer's environment.

Audit Trails

Full DAG run history, task logs, and execution metadata captured for regulatory reporting.

Access Controls

Role-based access and SSO integration enforce least-privilege across all pipeline operations.

Use Case 2: Large-Scale ETL/ELT Pipelines

The Challenge

Processing massive datasets across hybrid cloud environments demands reliable scheduling, fault-tolerant execution, and intelligent scaling. Complex transformations with data quality checks create dependency chains that must be monitored and alertable end-to-end.

The Astronomer Solution

Hosted Execution delivers simplicity and elastic scale for cloud-native pipelines. Remote Execution supports data locality requirements when large datasets cannot be moved. The Kubernetes Executor enables per-task resource isolation for heterogeneous workloads — from lightweight queries to heavy Spark or dbt transformations.

High-Throughput Scheduling

Handle thousands of concurrent task instances with the Celery Executor and autoscaling worker pools.

Data Quality Gates

Integrate dbt tests, Great Expectations, and custom sensors natively into DAG workflows.

Use Case 3: MLOps and Machine Learning Workflows

1

Data Preparation

Orchestrate feature engineering, dataset versioning, and preprocessing pipelines at scale.

2

Model Training

Schedule GPU-accelerated training jobs on Kubernetes, with dynamic resource allocation per workload.

3

Evaluation & Registry

Automate model evaluation, A/B testing gates, and registration to MLflow or SageMaker Model Registry.

4

Deployment & Monitoring

Trigger model deployments and schedule drift detection pipelines to keep models performant in production.

Use Case 4: Centralized Platform Governance

The Challenge

Large enterprises often have dozens of Airflow instances managed independently by different teams or business units. This creates inconsistency in security posture, Airflow version sprawl, duplicated infrastructure costs, and no cross-team visibility into pipeline health or resource consumption.

The Astronomer Solution

Astro's centralized control plane provides a single pane of glass across all Deployments and Workspaces. Platform teams enforce organization-wide standards — approved Airflow versions, resource limits, SSO policies — while individual teams retain full autonomy to build and deploy their own DAGs through self-service Workspaces.

- **Multi-tenancy** with namespace-level isolation between teams
- **Centralized RBAC** with SSO integration (Okta, Azure AD, Google)
- **Platform-wide observability** — resource usage, SLA tracking, alerting
- **Standardized tooling** via Astro CLI and shared CI/CD templates

Astronomer: Your Enterprise Airflow Advantage

Astronomer provides a secure, scalable, and fully-managed platform that transforms Apache Airflow from an operational challenge into a strategic business enabler.

Managed at Scale

Eliminate infrastructure toil. Astronomer manages upgrades, scaling, and reliability so your team ships pipelines faster.

Enterprise-Ready Security

Remote Execution, dedicated clusters, RBAC, and SSO give compliance and security teams full confidence.

Flexible Architecture

Hosted or Remote Execution, Standard or Dedicated Clusters — meet your data residency and network requirements without compromise.

Unified Governance

One control plane for all teams. Enforce standards centrally while empowering teams with self-service autonomy.

Ready to empower your enterprise with the future of data orchestration? Astronomer Managed Airflow is production-ready today.